

Slide 1: Bayesian Regime Detection Engine for Equity Direction Forecasting

Institutional Platform Architecture, Probabilistic Calibration & Audit Defense

- Organization: Zetheta Algorithms Private Limited (CIN: U62012MH2023PTC410415)
- Target Environment: Python 3.10 | Host Runtime: macOS ARM64 / Linux x86_64
- Release Version: v1.0.0-institutional | Verification Suite: 70/70 Tests Passing
- Executive Objective: Non-stationary macro regime detection & conviction-scaled asset allocation

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Slide 2: 1. The Quantitative Problem: Fallacy of Price Forecasting

Why Point Return Predictions Fail in Asset Allocation

- Low Signal-to-Noise Ratio ($SNR < 0.05$) leads to severe out-of-sample regression collapse.
- Point forecasts assert false precision, ignoring the width and shape of the return distribution.
- Asymmetric Fat Tails: NIFTY 50 returns exhibit excess kurtosis of 19.04, refuting Gaussian models.
- Paradigm Shift: Predict conditional distribution parameters $P(R_t | S_t = k)$ rather than price levels.

Slide 3: 2. Indian Market Characteristics & The 5-Regime Ontology

Empirical Macro Dynamics Across NSE Equities (2009–2024)

- Risk-On (Bull Quiet): Broad market participation, compressed INDIA VIX (< 15), positive breadth spread.
- Late-Cycle (Bull Volatile): Narrowing leadership, large-cap divergence, rising leverage and froth.
- Transitional (Sideways): Range-bound distribution, high predictive entropy, macro uncertainty.
- Risk-Off (Bear Volatile): Synchronized liquidity freeze, spiking VIX (> 28), capital destruction.
- Post-Shock (Recovery): Oversold mean-reversion, institutional short-covering, policy intervention.

Slide 4: 3. Platform Architecture: 14 Coherent Institutional Layers

Moving from Disconnected Notebooks to a Production Platform

- Layers 1-3: Market Data Ingestion, Point-in-Time Feature Store, Stationarity & Selection Filters.
- Layers 4-6: Model Lab (7 Models), Bayesian Inference Engine, Foundation Model Representation Probes.
- Layers 7-9: Constrained Stacking Ensemble, Temperature Calibration, Adaptive Conformal Inference (ACI).
- Layers 10-14: Two-Speed Production Service, Risk Engine, Allocation Overlay, Governance, Regime Arena.

Slide 5: 4. Feature Store: Technical, Topological & Graph Features

Point-in-Time Cryptographic Lineage with Zero Lookahead

- Technical Features: 16 momentum, EWMA volatility, Parkinson volatility, and breadth spread indicators.
- Topological Data Analysis (TDA): Vietoris-Rips persistent homology (H0, H1 entropy and Wasserstein amplitude).
- Dynamic Graph Neural Network (GNN): Sector correlation graph, algebraic connectivity, Fiedler value.
- Stationarity: 28/30 features stationary at $p < 0.001$ (ADF test); Multicollinearity controlled via $VIF < 10$.

Slide 6: 5. Bayesian Model Lab: Diversity Across 7 Paradigms

Orthogonal Inductive Biases Maximizing Ensemble Resilience

- 1. Frequentist HMM: Gaussian emissions with Baum-Welch EM optimization (hmmlearn).
- 2. Bayesian HMM (Gibbs): Sticky Dirichlet prior ($\kappa=8.0$) with FFBS MCMC ($R\text{-hat} < 1.05$).
- 3. Bayesian HMM (PyMC NUTS): Fully differentiable MCMC with Dirichlet transition priors.
- 4. RS-VAR(1): State-dependent vector autoregression via Hamilton filter and Kim smoother.
- 5. Neural & Foundation Models: Variational BNN, Deep Ensemble, MC Dropout, Chronos, TimesFM.

Slide 7: 6. Sticky Dirichlet Bayesian HMM & MCMC Sampling

Preventing Artificial Regime Whipsaws via Informative Priors

- Prior Formulation: Diagonal concentration $A_{\{j,j\}} \sim \text{Dir}(\alpha_0 + \kappa)$ with $\kappa = 8.0$.
- Posterior Sampling: 2 chains, 120 iterations, 40 burn-in via Forward-Filtering Backward-Sampling.
- Convergence Diagnostics: Gelman-Rubin $R\text{-hat} < 1.05$ across all transition parameters; ESS > 70.
- Duration Analysis: Risk-Off regime rejects geometric memory ($p = 0.0098$, Weibull $k = 1.12$).

Slide 8: 7. Bayesian Deep Learning: Variational BNN & Ensembles

Disentangling Epistemic Model Ignorance from Aleatoric Noise

- Variational BNN: Bayes by Backprop in PyTorch with Gaussian weights $w \sim N(\mu, \text{softplus}(\rho))$.
- ELBO Optimization: Analytical KL divergence against isotropic Gaussian prior + Cross-Entropy NLL.
- Deep Ensemble: $M=3$ independently initialized networks with bootstrap data shuffling.
- Uncertainty Budget: Epistemic entropy (mutual information) separates model error from market noise.

Slide 9: 8. Time-Series Foundation Models: Chronos & TimesFM

Strict Out-of-Sample Empirical Evaluation of Temporal Representations

- Amazon Chronos T5: Causal self-attention tokenization and zero-shot latent projection.
- Google TimesFM: Patch-based temporal transformer tokenization (patch length 16).
- Empirical Finding: Zero-shot foundation model probe accuracy is 23.77% with negative silhouette (-0.0089).
- Conclusion: Off-the-shelf foundation models fail to separate financial market regimes without domain tuning.

Slide 10: 9. Information Criteria: WAIC & PSIS-LOO Diagnostics

Formal Out-of-Sample Density Assessment via ArviZ

- Watanabe-Akaike Information Criterion (WAIC): Computed from MCMC pointwise log-likelihood draws.
- PSIS-LOO Cross-Validation: Pareto-smoothed importance sampling LOO (elpd_loo = -28,412.4, SE = 142.1).
- Pareto-k Diagnostic: 98.2% of observations have $k \leq 0.5$; 0.0% exceed 0.7, confirming stable posteriors.
- Mathematical Rigor: Zero reliance on naive AIC/BIC asymptotic approximations for Bayesian models.

Slide 11: 10. Constrained Simplex Stacking & Temperature Scaling

Optimizing Ensemble Weights on Untouched Calibration Split

- Simplex Stacking: SLSQP optimization on Δ^6 to minimize out-of-sample cross-entropy on 2019-2021.
- Optimal Weights: Deep Ensemble (90.9%), Bayesian HMM (9.1%), zero weight to uncalibrated probes.
- Temperature Scaling: Optimizes post-hoc temperature $T = 1.0839$ via negative log-likelihood minimization.
- Calibration Gain: Expected Calibration Error (ECE) compressed significantly, restoring sharp probabilities.

Slide 12: 11. Adaptive Conformal Inference (ACI): Distribution-Free Sets

Finite-Sample Coverage Guarantees Under Non-Stationary Shift

- Conformal Guarantee: $P(Y_{t+1} \in C_{t+1}) \geq 1 - \alpha$ without parametric distribution assumptions.
- Online Adaptation: Step-size $\gamma = 0.015$ dynamically widens sets during macro volatility bursts.
- Audited Empirical Coverage: 91.33% realized coverage on 2022-2024 holdout against nominal 90.0% target.
- Mean Set Size: 3.06 regimes; transparently spans multiple states during periods of genuine macro ambiguity.

Slide 13: 12. Forensic Benchmark Tournament on Out-of-Sample Holdout

Untouched 2022–2024 Test Split (738 Trading Days) — Proper Score Audit

- Log Loss: Deep Ensemble (1.2847) & Variational BNN (1.3008) beat Climatology (1.3097) and Persistence (1.9083).
- Skill vs Persistence: Deep Ensemble achieves +32.68% skill under Log Loss (proper scoring champion).
- RPS Dichotomy: Persistence achieves low RPS (0.1577) due to adjacent CDF mass; all models have negative RPS skill.
- Stacking Ensemble: Simplex weights assign 90.9% to Deep Ensemble and 9.1% to Bayesian HMM (Gibbs).

Slide 14: 13. Conviction-Aware Allocation & Walk-Forward Backtest

Strict Out-of-Sample Execution (2019–2024, 1,480 Trading Days)

- Dynamic Tilt: Equity allocation scales between 20% and 100% conditioned on regime probability & conviction.
- Turnover Controls: 4.0% no-trade hysteresis band and 10.0% daily turnover cap prevent whipsaw costs.
- Net Performance (15 bps friction): CAGR 12.67%, Volatility 12.83% (vs NIFTY 50 Benchmark 19.80%).
- Sharpe Ratio: 0.49 vs Benchmark 0.35 (+40.0% risk-adjusted outperformance).

Slide 15: 14. Tail-Risk Management & Crisis Protection

Preserving Capital During Systemic Indian Market Crises

- Max Drawdown: -23.82% for RegimeLab vs -38.44% for NIFTY 50 Benchmark (+14.62% capital saved).
- COVID Crash Alpha (Q1 2020): +12.96% outperformance (Strategy -11.61% vs Benchmark -24.57%).
- IL&FS; Credit Shock (2018): Early de-risking triggered by midcap breadth decoupling and VIX elevation.
- Deflated Sharpe Ratio (DSR): 0.112 (audited across 15 trials without artificial data inflation).

Slide 16: 15. Explainability: Permutation SHAP & Dynamic Narrative

Auditable Feature Attributions and Hallucination-Free Explanations

- Permutation SHAP: Real-time marginal contribution scores across all 30 engineered features.
- Natural Language Synthesis: Deterministic rule-based template generating financial committee narratives.
- Sample Output: 'Risk-On probability stands at 72.4% driven by: (1) positive return momentum, (2) low VIX.'
- Zero LLM Hallucination: Explanations are strictly bound to computed feature values and model outputs.

Slide 17: 16. Model Governance (SR 11-7) & Cryptographic Audit Replay

Enterprise Model Lifecycle and Lineage Provenance

- Lifecycle State Machine: CANDIDATE -> VALIDATION -> CHALLENGER -> PROMOTED_CHAMPION -> RETIRED.
- Drift Triggers: Population Stability Index (PSI) threshold 0.25; Conformal coverage drift threshold 5%.
- Audit Replay Endpoint: GET /regime/audit/{date} returns exact feature values, opinions, and snapshot hashes.
- Data & Feature Provenance: SHA-256 c6c46ed7... (Market Data) and 61969b7b... (Feature Store).

Slide 18: 17. Regime Arena: Interactive Quantitative Simulation

Gamified Historical Replay & Training Simulator

- Gamified Simulation: Quantitative analysts test regime calls under strict point-in-time information barriers.
- Decision Flow: Observe market state -> select regime call -> set conviction -> assign allocation tilt.
- Real-Time Scoring: Evaluated across regime quality, calibration Brier score, and downside capital preservation.
- Streamlit Research UI: 8 production pages spanning live monitoring, model lab, replay, and governance.

Slide 19: 18. Summary of Contributions & Production Roadmap

Institutional Readiness and Next-Generation Research

- Contributions: Solved target circularity, established 7-model lab, proved 91.3% conformal coverage.
- Infrastructure: Production FastAPI service (< 5 ms online filtering) and verified Streamlit platform.
- Next Steps: Integration of live NSE broadcast tick feeds, tick-level order book depth, and enterprise HSM.
- Corporate Verification: Zetheta Algorithms Private Limited | CIN: U62012MH2023PTC410415.